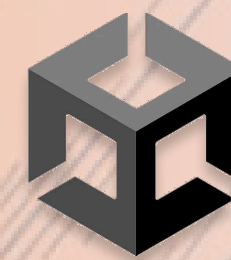




003

MOTE GUARDIAN



Lead Programmer & Designer

2024.08 - 2024. 10

This is a serious game designed to raise allergy protection awareness among preschoolers. I was responsible for the full project lifecycle, ranging from interaction prototyping to end-to-end software development.

On the technical side, I implemented a **Dynamic Difficulty Adjustment (DDA)** system using **LSTM neural networks**, enabling the game to adaptively tune challenge intensity based on each child's learning progress. By integrating cutting-edge AI with serious gaming, this project explores innovative applications of digital pedagogy within the field of children's health education.

DDA

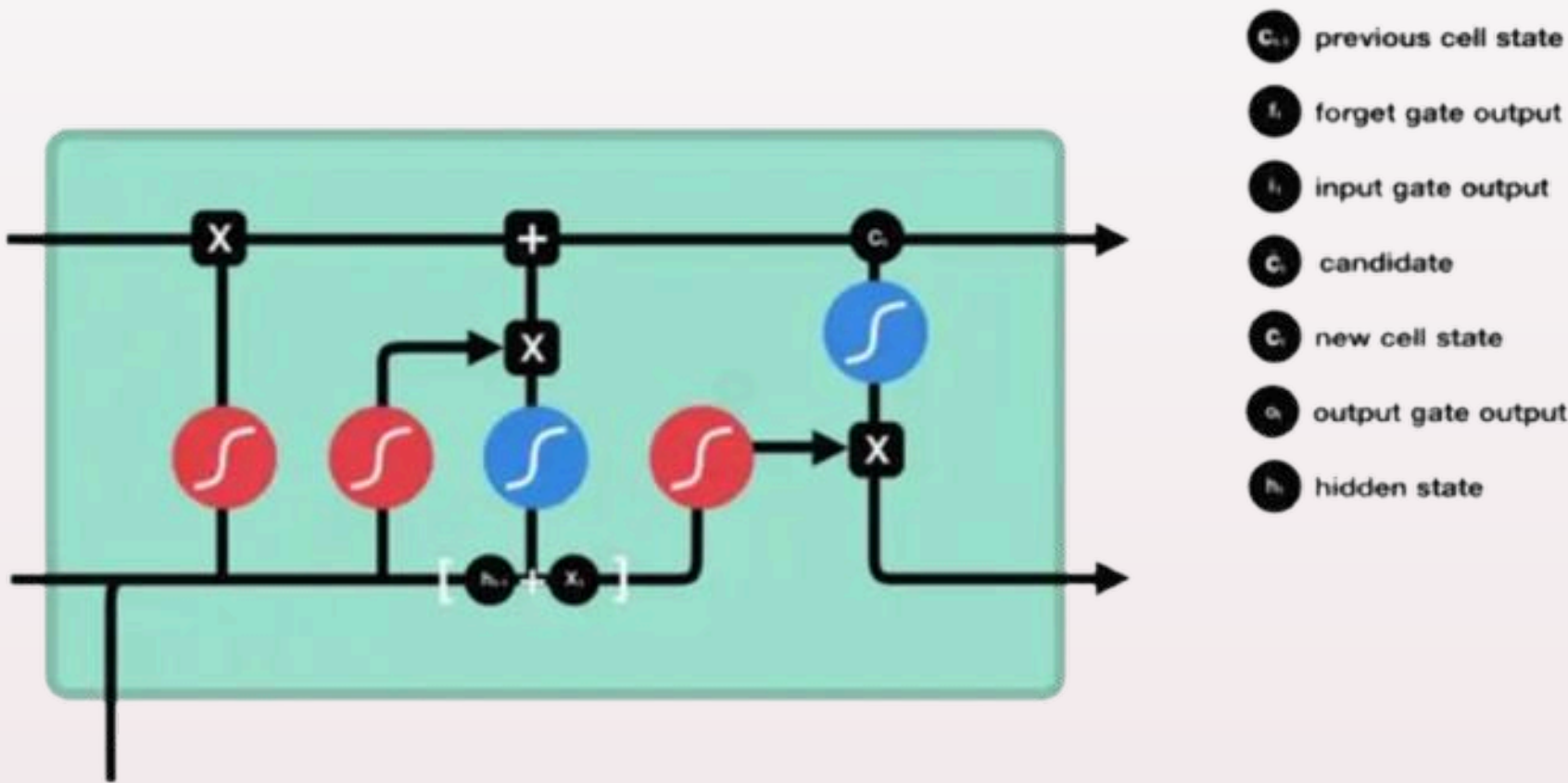
Preschoolers

LSTM

Background

As global allergy rates rise, preschoolers face the highest risks. Their limited cognitive maturity often leads to "accidental exposure" as they struggle to recognize allergens. Traditional education fails to capture the short attention spans of 3-8-year-olds.

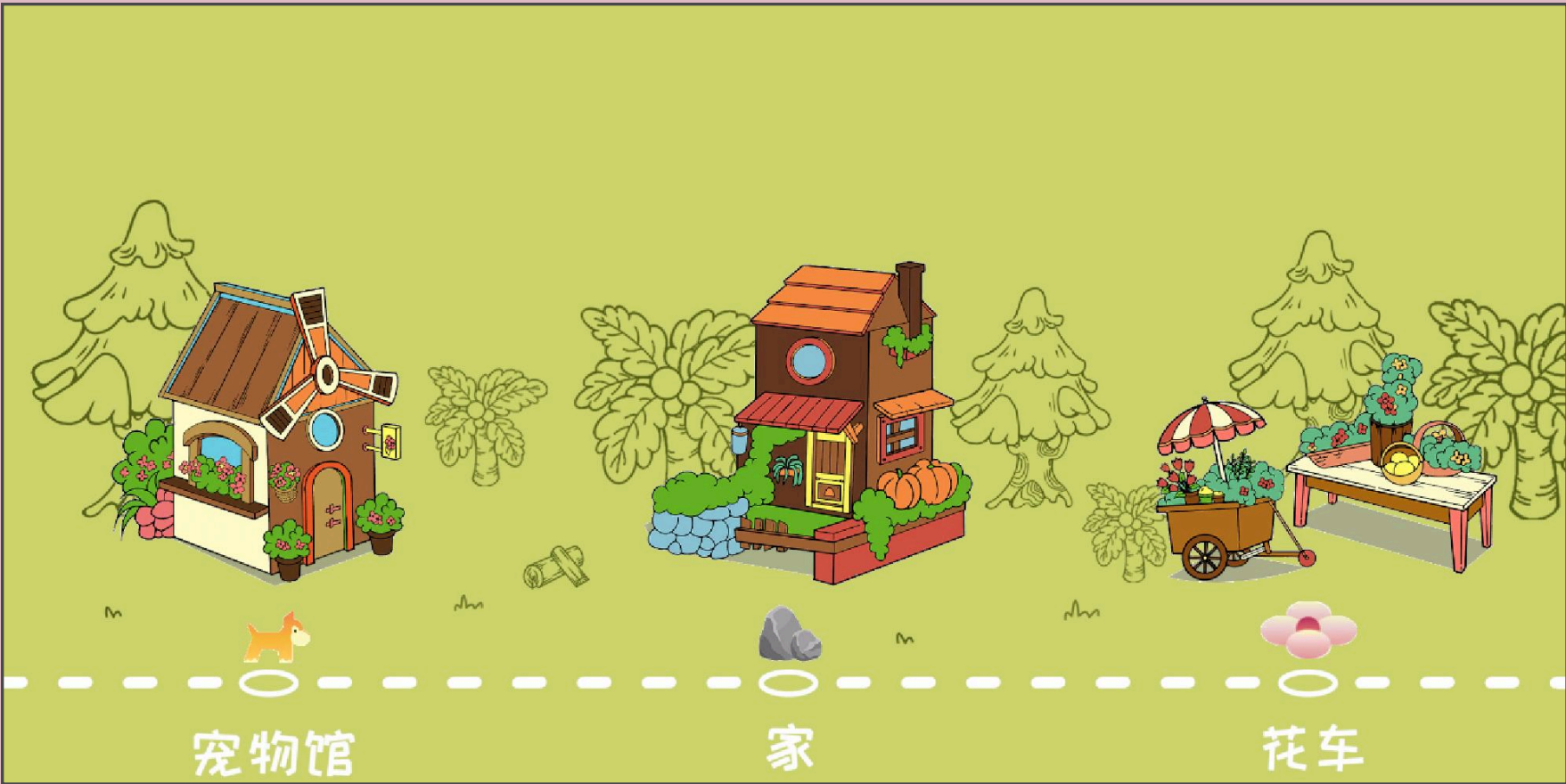
I developed **MOTE GUARDIAN** to bridge this gap. This serious game gamifies medical knowledge through interactive play. Its key feature is an LSTM-driven engine that analyzes player behavior in real-time to provide **Dynamic Difficulty Adjustment (DDA)**. This ensures the learning curve stays optimal (within the Zone of Proximal Development). My goal is to leverage AI and design to empower children with essential self-protection skills.



LSTM Working Principle

UI Design





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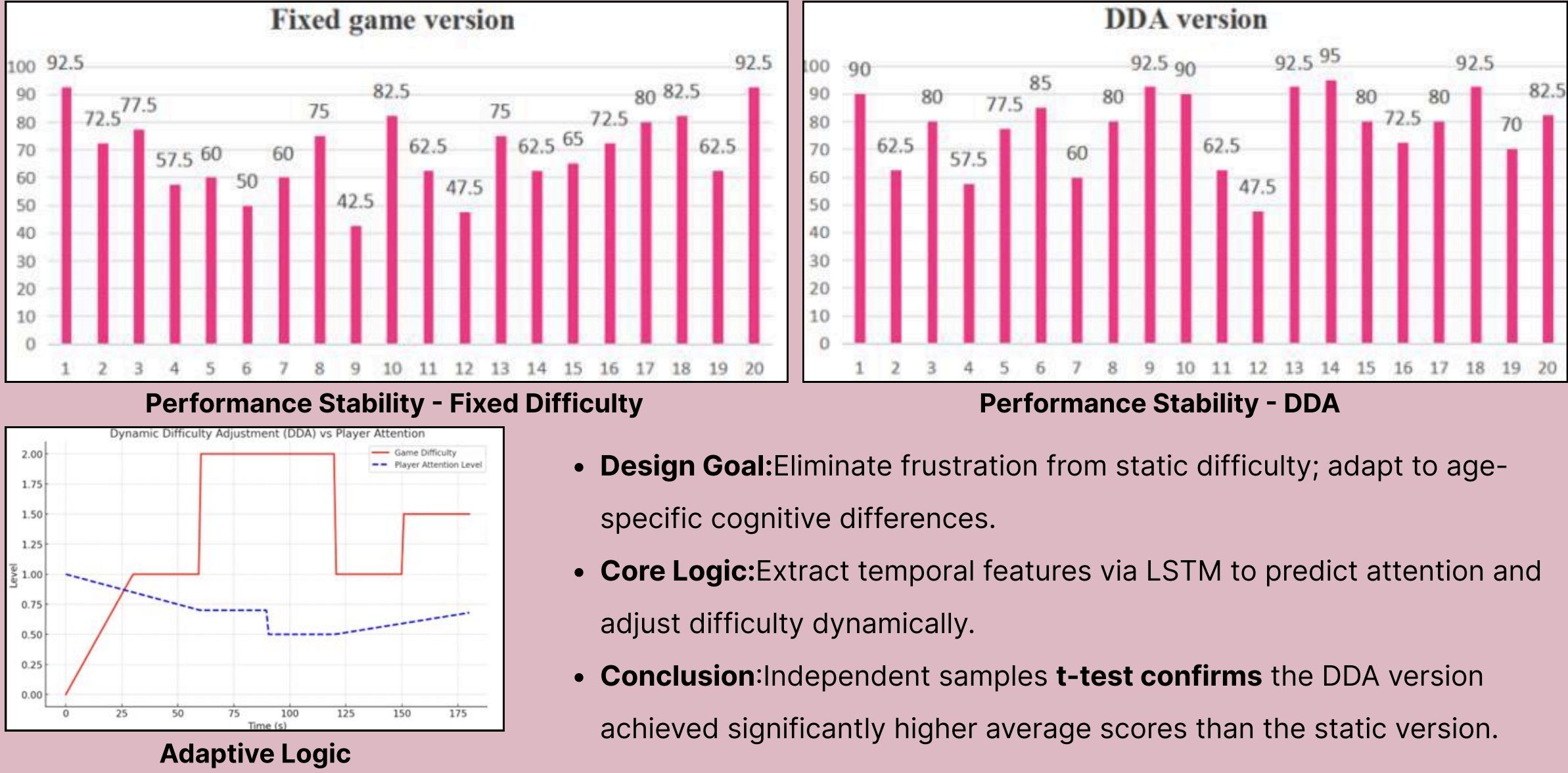
Players freely explore hand-drawn maps, triggering randomized allergen events. Visual cues are utilized to guide children in identifying potential hazards, while interactive mechanics—supplemented by engaging animations—reinforce their self-protection awareness.

Gameplay Flow

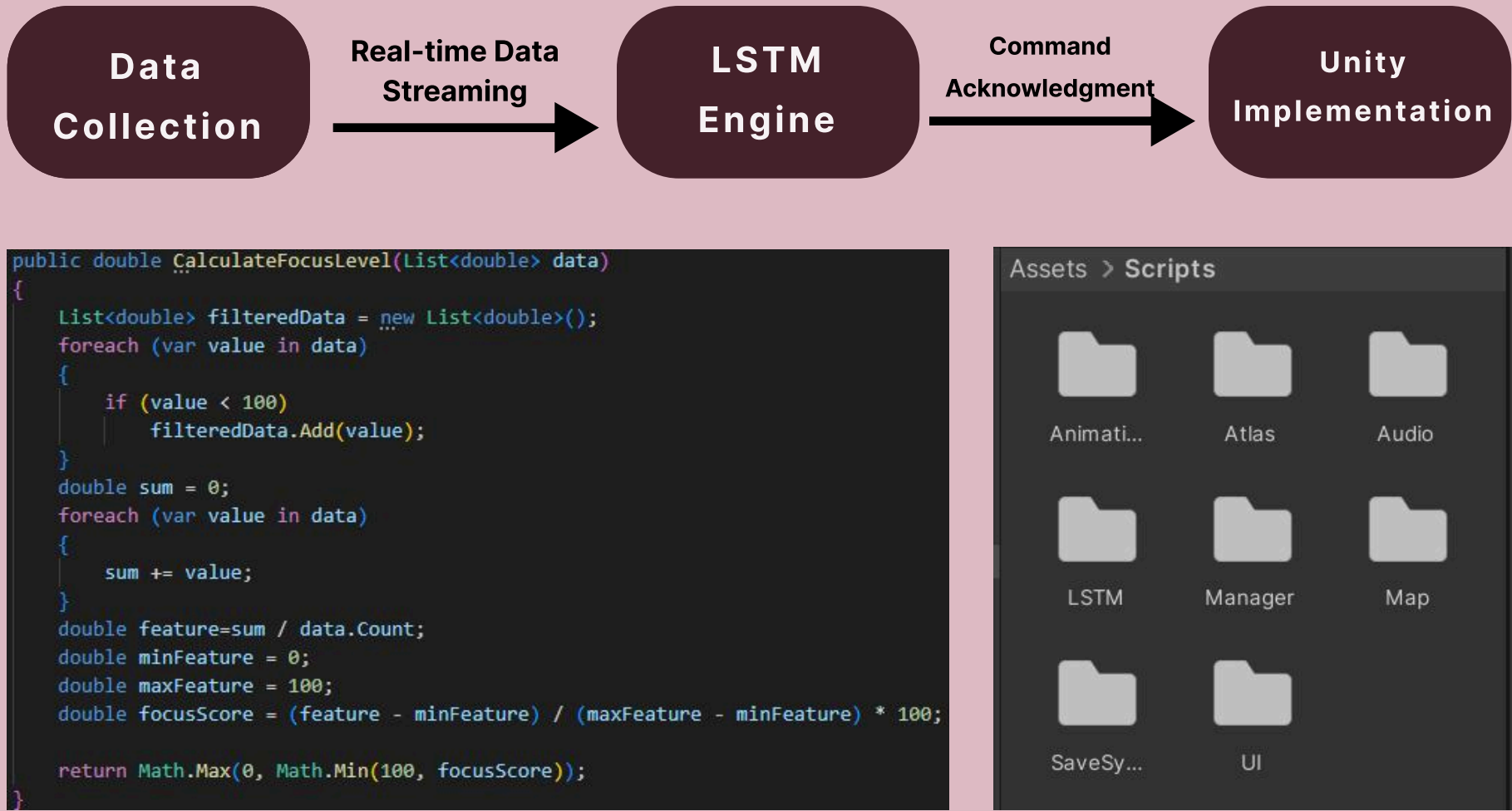


Technology & Solutions - 1

LSTM-Driven Intelligent Adaptation Framework



System Architecture & Unity Development



User Research & Analysis



Participants:
Recruited 10 children aged 3-8 and their parents to conduct usability testing.

Methodology:
Used **observational research** to record children's user flows, followed by **semi-structured interviews** with parents to evaluate educational effectiveness.

Feedback Mechanism:
DDA significantly extended average playtime and successfully maintained a state of "flow."

Technology & Solutions - 2

DDA Strategy



1 Escalating difficulty: Richer palettes, obscured recognition.



2 Escalating difficulty: Denser snowfall; rising snow levels



3 Escalating difficulty: Denser clouds and rain, obscured recognition.



4 Reduced difficulty: Enhanced visuals, improved clarity.